Portfolio — Hyeongju Lee

Data & Backend Engineer with a 3D computer-vision research background. Master's completion Mar. 2027.
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1. Data & Backend Engineering — Vision Inside (Jul. 2022 – Nov. 2024)

KISTI-commissioned Public R&D Value-Creation Platform (later SMART K2C) *Aug. 2022 – Jun. 2024*

- **Context:** an outsourcing contract from KISTI to build the data foundation of a platform that evaluates products' market competitiveness. University partners designed some of the models; Vision Inside built and ran the data and serving layer.
- **Role:** the project's **sole data engineer** on the Vision Inside side.
- **Collection:** Selenium crawlers for Amazon and Danawa and public-data-portal APIs for the Public Procurement Service, run on a 10–20-machine crawling farm feeding a central MySQL. Covered 10,985 of 11,046 Amazon categories (2,217,720 products, de-duplicated to 1,221,128 by ID/ASIN) and 2,481 Danawa categories. Later replaced Amazon scraping with the Rainforest API (selected 2023, implemented 2024).
- **Scale: 80M+ rows** by Nov. 2023 — Amazon 0.93M products / 10M detail rows / 7.9M review keywords; Danawa 0.42M products / 16M reviews; procurement 4.8M records / 42M attribute rows.
- **Models in production:** price-band K-means clustering with Dunn-index k selection (competitor benchmarking); a daily KeyBERT + Sentence-Transformers + Mecab batch extracting customer-required quality keywords from reviews; SQL matching of NTIS national R&D projects to NICE company records (248K links); procurement price/attribute/delivery statistics in SQL for the 2024 model set.
- **Serving:** FastAPI services with Jinja2 + Tabulator admin dashboards (filtering, sequential batch runs, unified search, Excel export); migrated the partner-built prototype (models 1–2, Java/Tomcat web app) to company servers and set up its Docker runtime.
- **Delivery:** wrote the audit deliverables (operator manual, initial data-build report); presented crawling and model results in technical meetings with KISTI researchers and handled their data and code requests directly. The project later led to a KRW 300M main contract.
- **Stack:** Python, Selenium, BeautifulSoup, pandas, SQLAlchemy, MySQL, scikit-learn, KeyBERT, Sentence-Transformers, FastAPI, Jinja2, Tabulator. Details: zoocasso.github.io/en/work/kisti

Genomic Variant Annotation & Risk-Report System (Kyung Hee Univ. × Canadian bio lab) *Aug. 2022 – Sep. 2023*

- **Problem:** in an international collaboration between Kyung Hee University's College of Korean Medicine and a Canadian bio research lab, turn customer VCF or chip (genotyping-array) input into readable disease/trait reports using large public genomic references.
- **Role: backend and pipeline developer** (algorithm designed with the team lead; a separate developer built the frontend).
- **Variant calling:** FASTQ → BWA → samtools → Picard → GATK → VCF; PyVCF + regex normalization of heterogeneous VCF/array inputs.
- **Reference data (∼1 TB):** dbSNP GRCh37/38 (237 GB), 1000 Genomes (770 GB), GWAS Catalog, UniProt variants; loaded into MySQL via chromosome-wise splitting (sed/split) and chunked pandas/SQLAlchemy ingestion. Annotation is a 4-stage SQL join keyed on position and alleles or rsID.
- **Report logic:** GWAS odds-ratio/beta-based risk scores per disease/trait, allele-frequency comparison across 26 1000 Genomes populations, and drug/herbal-medicine lookups (MedlinePlus, MeSH, ATC, PharmDB-K, OASIS), exposed as a FastAPI JSON API. LDA and 1D-CNN disease-to-herb models were prototyped.
- **Deployment:** Docker images through a Harbor private registry I built on a Korea IDC Ubuntu server, with domain and HTTPS (Let's Encrypt); access restricted to internal IPs (not publicly launched).
- **Collaboration:** technical bridge between the Korean and Canadian teams, working with the lab's designer.
- **Stack:** Python, BWA, samtools, Picard, GATK, PyVCF, pandas, SQLAlchemy, MySQL, FastAPI, Docker, Harbor. Code (rewritten Oct. 2026): github.com/zoocasso/bio_fastapi. Details: zoocasso.github.io/en/work/genomics

Life-Insurance Disclosure Data Tooling (team of three) *Aug. – Oct. 2023*

- **My part:** loaded public life-insurance disclosure workbooks (16 product categories) into MySQL with openpyxl, pandas, and SQLAlchemy; built a FastAPI + Tabulator web viewer; packaged xlwings-based VBA execution as a standalone tool; built a PyQt desktop prototype (finplot, PyInstaller); extracted a VBA pricing model's named ranges into MySQL.
- **Stack:** Python, openpyxl, xlwings, pandas, SQLAlchemy, MySQL, FastAPI, PyQt, PyInstaller.

2. Research & Applied Vision — Kyushu Institute of Technology (Apr. 2025 – present)

Training-free Light-Field / Integral-Imaging Reconstruction (advisor: Prof. Min-Chul Lee) *4 first-author papers*

- **Ray-based Inverse Mapping (RIM):** redistributes angular samples to spatial positions as a bijection with zero quantization error under orthographic conditions. On 7 Stanford Lytro scenes: PSNR 23.31 → 25.70 dB (**+2.39 dB**), SSIM 0.741 → 0.877;

downstream YOLO detection mAP50 0.665 → 0.776 (**+0.11**). Variant iRIM synthesizes virtual lens arrays from multi-view images. Details: zoocasso.github.io/en/research/rim

- **Accelerated origin-point optimization:** replaced the iterative search with direct camera-parameter computation — **81 reconstructions → 1** on 9×9 elemental images (paper: 0.31 s on CPU vs. an estimated 25.45 s = 81 × 0.3142 s; 0.0057 s on GPU with CuPy, RTX 3090); same origin and SSIM in the paper's experiments. Details: zoocasso.github.io/en/research/origin-point
- **Implementation:** Python, NumPy, CuPy; evaluation loop with PSNR, SSIM, and detection mAP.

Image-Data Pipeline — commissioned research with Hitachi *Apr. 2025 – present*

- **Role:** research assistant (part-time) in a Kyushu Institute of Technology lab.
- **Problem:** the inherited code for producing training images could not be maintained, and dataset creation depended on developers.
- **Built:** an end-to-end Python pipeline — machine-vision camera control and capture, cleaning, preprocessing into model-ready datasets — with a PyQt GUI that non-developer researchers operate themselves.
- **Outcome:** deployed in a lab in the Netherlands: used hands-on by Hitachi staff, demonstrated to the Dutch researchers. Design handed over to two junior researchers. Working languages: Japanese and English.
- **Stack:** Python, OpenCV, Spinnaker SDK (PySpin), PyQt, PyInstaller. Details: zoocasso.github.io/en/work/hitachi

3. Other Projects

- **3DOS — vision enhancement on an XR headset** (2024 XR Device Content Makerthon finals, Nov. 2024; team 3DOS, five lab members registered) — **sole developer and live demo**. Fourier peplography (fog/smoke removal) and photon counting (low-light recovery), methods from Prof. Myungjin Cho's research, in C++/OpenCV, built as an Android native plugin (arm64) and loaded by a Unity (IL2CPP) app on the Metalens 2 headset, applied to its own camera feed. Code lost. Slides: github.com/zoocasso/realtime-vision-xr-device
- **WJMax game server** (side project, team of two, Aug. 2023 – Feb. 2024). Sole server developer: FastAPI on Docker (AWS EC2) with RDS MySQL behind a Cloudflare proxy/WAF, and a play-log ingestion API for a community rhythm game (∼300 average, ∼2,000 peak concurrent players during updates). Code (rewritten Oct. 2026): github.com/zoocasso/wjmax-game-server; details: zoocasso.github.io/en/work/wjmax
- **Smart Mirror** (capstone, team of two, 2022). Raspberry Pi; OpenCV LBPH face recognition and Kakao speech-to-text voice control for per-user dashboards on a half-mirror display.
- **Depth estimation via correlation filter + VCR** (course project, 2022, MATLAB). Built on the course's VCR and correlation-filter scripts, applied to the Three Characters set (small 2022 changes; rewritten Oct. 2026) — the starting point of the master's research.
- **CNN in MATLAB** (self-study, 2022). MNIST CNN (convolution, ReLU, max pooling, softmax) with backpropagation and momentum SGD, following Phil Kim's *MATLAB Deep Learning*.

4. Publications (first author)

- Hyeongju Lee, Myungjin Cho, Min-Chul Lee, "Angular-to-Spatial Information Redistribution via Ray-based Inverse Mapping," *ICMV 2026*, Budapest (poster, Best Poster Award).
- Hyeongju Lee, Myungjin Cho, Min-Chul Lee, "Virtual Lens Array Synthesis from Multi-View Images via iRIM," *ICCAS 2026*, Sapporo (poster).
- Hyeongju Lee, Myungjin Cho, Min-Chul Lee, "Accelerated Origin Point Optimization for Volumetric Computational Reconstruction," *ITC-CSCC 2026*, Bangkok (oral). Code for both ITC-CSCC papers: github.com/zoocasso/origin-point-optimization-vcr
- Hyeongju Lee, Gilsu Yeo, Myungjin Cho, Min-Chul Lee, "Image Enhancement of Volumetric Computational Reconstruction Using Origin Point Optimization," *ITC-CSCC 2025*, Seoul (poster).
- Hyeongju Lee et al., "Image Enhancement of Volumetric Computational Reconstruction Using Origin Point Optimization" (in Korean), *Photonics Conference 2024*, Optical Society of Korea (poster).

5. Skills, Education & Recognition

- **Skills:** Python, SQL, FastAPI, SQLAlchemy, pandas, MySQL, Selenium, Docker, Harbor, Linux, AWS (EC2, RDS), Cloudflare; PyTorch, scikit-learn, OpenCV, CuPy, KeyBERT; BWA, samtools, Picard, GATK; PyQt, openpyxl, xlwings; C++, Java, JavaScript, C#, MATLAB (basic–intermediate).
- **Education:** Master's program, Computer Science and Systems Engineering (Creative Informatics), Kyushu Institute of Technology, Apr. 2025 – Mar. 2027 (expected), GPA 3.63/4.0. B.S., Electrical, Electronic & Control Engineering, Hankyong National University, Feb. 2025, GPA 4.00/4.50.
- **Recognition & other:** ICMV 2026 Best Poster Award (Oct. 2026); Rotary Yoneyama Memorial Foundation Scholar (2026, Japanese-language interview); SQLD — SQL Developer (Oct. 2023); military service completed as Industrial Technical Personnel (Jul. 2022 – Jul. 2024). Languages: Korean (native), Japanese (business conversational), English (working proficiency).